

## *MTH302- Business Mathematics & Statistics*

| <u>Paper</u>      | <u>Year</u> | <u>Session</u>      |
|-------------------|-------------|---------------------|
| <i>Paper # 01</i> | <i>2008</i> | <i>(session-02)</i> |
| <i>Paper # 02</i> | <i>2009</i> | <i>(session-02)</i> |
| <i>Paper # 03</i> | <i>2009</i> | <i>(session-03)</i> |
| <i>Paper # 04</i> | <i>2009</i> | <i>(session-04)</i> |
| <i>Paper # 05</i> | <i>2010</i> | <i>(unknown)</i>    |
| <i>Paper # 06</i> | <i>2010</i> | <i>(unknown)</i>    |
| <i>Paper # 07</i> | <i>2011</i> | <i>(session-02)</i> |
| <i>Paper # 08</i> | <i>2011</i> | <i>(session-03)</i> |
| <i>Paper # 09</i> | <i>2011</i> | <i>(session-04)</i> |

### *Paper # 01* *Fall 2008* *(Session - 2)*

**Question No: 1 ( Marks: 1 ) - Please choose one**

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Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10 . The % Markdown is

- ▶ 10%
- ▶ **20%**
- ▶ 30%
- ▶ 40%

**Question No: 2 ( Marks: 1 ) - Please choose one**

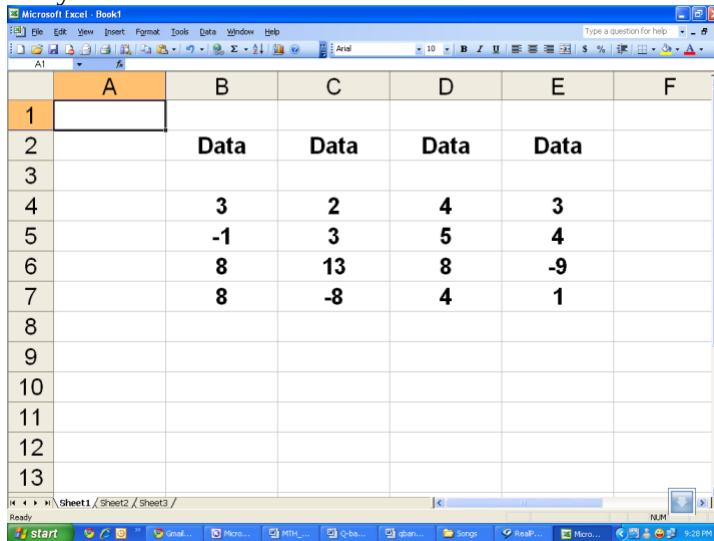
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Given Net price of shirt = \$20 , Discount = 10% ,List price will

- ▶ \$22
- ▶ \$24
- ▶ \$26
- ▶ \$30

**Question No: 3 ( Marks: 1 ) - Please choose one**

Which is the correct syntax for the determinant of a matrix given by the following array



|    | A | B    | C    | D    | E    | F |
|----|---|------|------|------|------|---|
| 1  |   |      |      |      |      |   |
| 2  |   | Data | Data | Data | Data |   |
| 3  |   |      |      |      |      |   |
| 4  |   | 3    | 2    | 4    | 3    |   |
| 5  |   | -1   | 3    | 5    | 4    |   |
| 6  |   | 8    | 13   | 8    | -9   |   |
| 7  |   | 8    | -8   | 4    | 1    |   |
| 8  |   |      |      |      |      |   |
| 9  |   |      |      |      |      |   |
| 10 |   |      |      |      |      |   |
| 11 |   |      |      |      |      |   |
| 12 |   |      |      |      |      |   |
| 13 |   |      |      |      |      |   |

- ▶ =DETERM (B4:E7)
- ▶ =MDTERM (B4:E7)
- ▶ **=MDETERM (B4:E7)**
- ▶ =MDETERM (B4;E7)

**Question No: 4 ( Marks: 1 ) - Please choose one**

**VDB** returns the depreciation of an asset for

- ▶ Zero period
- ▶ One period
- ▶ Two periods
- ▶ **Any arbitrary period**

**Question No: 5 ( Marks: 1 ) - Please choose one**

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%
- ▶ **25%**
- ▶ 20%
- ▶ 10%

**Question No: 6 ( Marks: 1 ) - Please choose one**

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$8,000. The useful life of the asset is 5 years. The depreciation is being calculated for the third year, and there are 10 months in the first year.

- ▶ =DB (10000, 5, 8000, 3, 10)
- ▶ =DB (10000, 8000, 5, 10, 3)
- ▶ **=DB (10000, 8000, 5, 3, 10)**
- ▶ =DB (10000, 8000, 10, 3, 5)

**Question No: 7 ( Marks: 1 ) - Please choose one**

Which of the following functions is not related to financial analysis?

- ▶ AMORDEGRC
- ▶ AMORLINC
- ▶ CUMIPMT
- ▶ **MDETERM**

**Question No: 8 ( Marks: 1 ) - Please choose one**

(2)  $\wedge -2 = )$  **0.25.....**

- 

**Question No: 9 ( Marks: 1 ) - Please choose one**

□ Dimension of a matrix means

□□□□□□▶ □ Column space

□□□□□□▶ □ Row space



► Simple interest method

► **Compound interest method**

► Annuity

► Hit & trial method

**Question No: 15 ( Marks: 1 ) - Please choose one**

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Suppose that you are spending 1500 dollars in an investment and Accumulated factor is 15 then Accumulated Value will be

► 22550

► **22500**

► 21000

► 22510

**Question No: 16 ( Marks: 1 ) - Please choose one**

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$3\frac{1}{2}$

What is the interest on Rs. 1600 for one year at the rate %?

► 65

► **56**

► 75

► 90

**Question No: 17 ( Marks: 2 )**

---

A man borrows \$39000 for 1 and half year at a rate of 1.65%. What is the simple interest he has to pay?

Simple interest is given by

$$I = \frac{P \times R \times T}{100}$$

P= principal = \$39000

R= interest rate= 1.65%

T= 1.5 year

$$\text{Interest, } I = \frac{(39000 \times 1.65 \times 1.5)}{100} = \text{Rs } 965.25$$

**Question No: 18 ( Marks: 2 )**

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If

Cost price = 2000Rs

Selling price = 6500Rs

Then what will be %Markup on cost?

MUC= S- C (Rs markup on cost = selling price- cost price)

= 6500-2000 = Rs 4500

% markup on cost = (MuC/cost price) \* 100 = (4500/2000) \*100= 225%

**Question No: 19 ( Marks: 3 )**

|    | A | B                 | C |
|----|---|-------------------|---|
| 1  |   |                   |   |
| 2  |   |                   |   |
| 3  |   | 50                |   |
| 4  |   | 20                |   |
| 5  |   | 10                |   |
| 6  |   | 2                 |   |
| 7  |   |                   |   |
| 8  |   |                   |   |
| 9  |   | =SUM(B3,B4,B5,B6) |   |
| 10 |   | =MULTIPLY(B5,B6)  |   |
| 11 |   | B4/B6             |   |
| 12 |   | =B4%              |   |
| 13 |   |                   |   |

Write the output of B9, B10, B11 and B12. If you think the formula is incorrect write "Error" in output.

**ANSWER:**

The output of =sum(B3,B4,B5,B6) = 82

=multiply(B5,B6) will not give result because multiply is not a excel function

B4/B6 is not a formula because in excel formula must start with "="

=B4% will return 0.2

**Question No: 20 ( Marks: 5 )**

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A gold chain is sold for Rs. 6500 at a gain of 25%. Find the profit.

Selling price = 6500

Profit = 25%

Profit in Rs= (6500\*25)/100= Rs 1625

**Question No: 21 ( Marks: 10 )**

---

For the following matrices perform the indicated operation, if possible.

$$A = \begin{bmatrix} 2 & 0 & -3 & 2 \\ -1 & 8 & 10 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -4 & -7 & 2 \\ 12 & 3 & 7 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 & 2 \\ -4 & 9 & 5 \\ 6 & 0 & -6 \end{bmatrix}$$

(a)  $A + B$

(b)  $B - A$

(c)  $A + C$

a)

$$A = \begin{matrix} 2 & 0 & -3 & 2 \\ -1 & 8 & 10 & -5 \end{matrix} \quad + \quad B = \begin{matrix} 0 & -4 & -7 & 2 \\ 12 & 3 & 7 & 9 \end{matrix}$$

$$A+B = \begin{matrix} 2+0 & 0-4 & -3-7 & 2+2 \\ -1+12 & 8+3 & 10+7 & -5+9 \end{matrix}$$

$$A+B = \begin{matrix} 2 & -4 & -10 & 4 \\ 11 & 11 & 17 & 4 \end{matrix}$$

b)

$$B-A = \begin{matrix} 0-2 & -4-0 & -7-(-3) & 2-2 \\ 12-(-1) & 3-8 & 7-10 & 9-(-5) \end{matrix}$$

$$B-A = \begin{matrix} -2 & -4 & -4 & 0 \\ 13 & -5 & -3 & 14 \end{matrix}$$

c)  $A+C$  is not conformable for addition because for addition number of rows and columns of A must be equal to number of rows and columns of C

***Paper # 02***  
***Fall 2009 (Session - 2)***

***Time: 60 min***  
***Marks: 42***

**Question No: 1 ( Marks: 1 ) - Please choose one**

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45% of what is 9?

- ▶ 20
- ▶ 40
- ▶ 30
- ▶ 10

**Question No: 2 ( Marks: 1 ) - Please choose one**

|    | A | B          | C          | D                        | E | F | G | H | I | J |
|----|---|------------|------------|--------------------------|---|---|---|---|---|---|
| 1  |   |            |            |                          |   |   |   |   |   |   |
| 2  |   |            |            |                          |   |   |   |   |   |   |
| 3  |   |            | List price | Discounted Price         |   |   |   |   |   |   |
| 4  |   |            |            |                          |   |   |   |   |   |   |
| 5  |   | Calendar   | 100        | 90                       |   |   |   |   |   |   |
| 6  |   | Pen        | 150        | 130                      |   |   |   |   |   |   |
| 7  |   | Diary      | 500        | 470                      |   |   |   |   |   |   |
| 8  |   | Table Lamp | 400        | 385                      |   |   |   |   |   |   |
| 9  |   |            |            |                          |   |   |   |   |   |   |
| 10 |   |            | SUMIF =    | =SUMIF(C5:C8,>110,D5:D8) |   |   |   |   |   |   |
| 11 |   |            |            |                          |   |   |   |   |   |   |
| 12 |   |            |            |                          |   |   |   |   |   |   |
| 13 |   |            |            |                          |   |   |   |   |   |   |
| 14 |   |            |            |                          |   |   |   |   |   |   |
| 15 |   |            |            |                          |   |   |   |   |   |   |
| 16 |   |            |            |                          |   |   |   |   |   |   |
| 17 |   |            |            |                          |   |   |   |   |   |   |
| 18 |   |            |            |                          |   |   |   |   |   |   |

What will be the answer of the SUMIF formula?

- ⌚ 1075
- ⌚ 985
- ⌚ 855
- ⌚ 130

**Question No: 3 ( Marks: 1 ) - Please choose one**

After the merchant buys merchandise, it is sold at a higher price called the \_\_\_\_\_

- ▶ Sale price
- ▶ Revenue discount
- ▶ Selling price
- ▶ Cost price

**Question No: 4 ( Marks: 1 ) - Please choose one**

In the case of partial payments, the \_\_\_\_\_ is the partial payment plus the partial cash discount

- ▶ Amount received
- ▶ Amount returned
- ▶ Amount credited
- ▶ Amount displaced

**Question No: 5 ( Marks: 1 ) - Please choose one**

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In a friendly cricket between India and Pakistan, Pakistan is scoring at a rate 7% more than India in 35 overs. If the India's score after 35 overs was 230 what's Pakistan score now?

- ▶ **250**
- ▶ **235**
- ▶ **246**
- ▶ **240**

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

0.20% = -----

□□□□□□□

▶ □0.2

⌚ 0.02

⌚ **0.002**

⌚ 0.0002

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

$$A = \begin{bmatrix} 2 & 6 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix}$$

, then AB is -----

$$\begin{bmatrix} 20 & 36 \\ 25 & 30 \end{bmatrix}$$

▶

$$\begin{bmatrix} 20 & 36 \\ 35 & 30 \end{bmatrix}$$

▶

$$\begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$$

▶

---

$$\begin{bmatrix} 20 & 36 \\ 45 & 39 \end{bmatrix}$$

▶

**Question No: 8 ( Marks: 1 ) - Please choose one**

What will be the base of 25% rate and percentage is 65?

- ▶ 216
- ▶ 250
- ▶ 260
- ▶ None of these

**Question No: 9 ( Marks: 1 ) - Please choose one**

The formula for compound interest for n number of periods is

☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐  $S = P(1 + r/100)^* n$

☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐  $S = P(1 + n/100)^{\wedge} r$

☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐  $S = P(1 + r/100)^{\wedge} n$

☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐  $S = P(1 + r/100)/ n$

**Question No: 10 ( Marks: 1 ) - Please choose one**

According to the local laws, the allowances are considered tax free if they are up to

- ▶ half of the basic salary
- ▶ one fourth of the basic salary
- ▶ one third of the basic salary
- ▶ 55% of the basic salary

**Question No: 11 ( Marks: 1 ) - Please choose one**

*Amount of discount is obtained as*

☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Percentage of Discount x List Price

- ☐ Percentage of Discount / List Price
- ☐ Percentage of Discount - List Price
- ☐ None of these

**Question No: 12 ( Marks: 1 ) - Please choose one**

In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- **Gratuity Fund**

**Question No: 13 ( Marks: 1 ) - Please choose one**

The value of x for the equation  $-(x - 5) = x - 2$  is -----

► no solutions

- ▶ -5
- ▶ 3
- ▶  $1/2$

**Question No: 14 ( Marks: 1 ) - Please choose one**

What will be the simple interest earned on an amount of Rs. 16,800 in 9 months at the

$$6\frac{1}{4}$$

rate of % p.a?

□ □ □ □ □ □ □ ▶ □ **Rs. 787.50**

□ □ □ □ □ □ □ ▶ □ Rs. 812.50

□□□□□□▶ □Rs 860

Rs. 887.50

**Question No: 15 ( Marks: 1 ) - Please choose one**

This example returns the present value of an investment that pays Rs. 250 at the end of every month for 2 years. The money paid out will earn 7.5% annually.

□□□□□□□▶□=PV (7.5%/12, 2\*12, 250, 1)

□□□□□□▶□=PV (7.5%/12, 2\*12, 250, 0)

□□□□□□▶□=PV (7.5%/12, 2, 250, 0)

□□□□□□▶□=PV (7.5%, 2\*12, 250, 1)

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

All formula in Excel start with

[illegible]

[illegible]

**Question No: 17 ( Marks: 1 ) - Please choose one**

A percentage is a way of expressing a number as



► □ a fraction of 100.

□□□□□□▶ □sum of number with 100.

□□□□□□▶ □multiplication of number with 100.

□□□□□□□▶ □average of number with 100.

**Question No: 18 ( Marks: 1 ) - Please choose one**

Given : List Price = 5500Rs , Discount = 850Rs, then Net Cost Price will be

□□□□□□▶ □6350 Rs

6350 Rs

□□□□□□▶ □5508 Rs

5508 Rs

□□□□□□▶ □5585 Rs

□ □ □ □ □ □ □ ▶ □ 4650 Rs

**Question No: 19 ( Marks: 1 ) - Please choose one**

If C = Cost price, S = Selling Price then the margin on the cost is equal to

$$\frac{C - S}{C} \times 100$$

$$\frac{C - S}{S} \times 100$$

□ □ □ □ □ □ □ ▶ □

□ □ □ □ □ □ □ ▶ □

 The correct answer cannot be determined without more information

**Question No: 21 ( Marks: 1 ) - Please choose one**

The net price of a computer table is 2500 and list price 3000. The trade discount will be

► 13%

► **17%**

► 21%

► 25%

**Question No: 22 ( Marks: 1 ) - Please choose one**

Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10

The % Markdown is

► 10%

► 20%

▶ 30%

▶ 40%

**Question No: 23 ( Marks: 1 ) - Please choose one**

---

Given for a product selling price = \$45 and unit variable cost= \$30 then Unit contribution margin will be

▶ \$15

▶ \$75

▶ \$1.5

▶ ***Can not be found from given values.***

**Question No: 24 ( Marks: 1 ) - Please choose one**

---

$3x^2 + 5x - 7$  is ----- expression.

▶ Monomial

▶ ***Binomial***

▶ Trinomial

▶ Linear

**Question No: 25 ( Marks: 3 )**

---

The mean of the numbers 10, 7, 9, 2 and x is 11; find the value of x.

Solution

total =  $5 \times 11 = 55$

total =  $10 + 7 + 9 + 2 + x = 55$

$28 + x = 55$

$x = 55 - 28$

$x = 27$

**Question No: 26 ( Marks: 5 )**

---

The salary of an employee is as follows:

Basic salary = 50,000 Rs.

What is the amount of allowances if House Rent = 45 %, Conveyance allowance = 2.5 % and Utilities allowance = 2.5 %?

Solution

$$\text{House Rent} = 50000 \times 45/100 = 22500$$

$$\text{Conveyance Allowance} = 50000 \times 2.5/100 = 1250$$

$$\text{Utilities Allowance} = 50000 \times 2.5/100 = 1250$$

**Question No: 27 ( Marks: 10 )**

If the price of 16 dozen of eggs and 10 breads of large size is 332. The price of 10 dozen of eggs and 8 breads of large size is 225. Find the prices of per dozen eggs and per bread.

(Solve this question by using matrices only)

Solution

$$192x + 10y = 332$$

$$120x + 8y = 225$$

from 2

$$8y = 225 - 120x$$

$$y = (225 - 120x)/8$$

put in 1

$$192x + 10(225 - 120x)/8 = 332$$

$$192x + 281.25 - 150x = 332$$

$$(192 - 150)x = 332 - 281.25$$

$$42x = 50.75$$

$$x = 1.21$$

$$y = (225 - 120 \times 1.21)/8$$

$$y = 10$$

***Paper # 03***  
***Fall 2009 (Session - 3)***

***Time: 60 min***  
***Marks: 42***

**Question No: 1 ( Marks: 1 ) - Please choose one**

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The Basic salary of an employee is Rs 7,000. What is the contribution of the company on account of gratuity to the Gratuity Trust Fund?

- ▶ **Rs 636.36**
- ▶ Rs 6363.6
- ▶ Rs 63.636
- ▶ Rs 6363

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

The monthly incomes of A and B are in the ratio 4 : 5, their expenses are in the ratio 5 : 6. If 'A' saves Rs.25 per month and 'B' saves Rs.50 per month, what are their respective incomes?

- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Rs.400 and Rs.500
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☒ **Rs.240 and Rs.300**
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Rs.320 and Rs.400
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Rs.440 and Rs.550

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

Sum of annuity is always

- ▶ Present value
- ▶ **Future value**
- ▶ Net present value
- ▶ Current value

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

In annuity interest is charged by the:

- ▶ Simple interest method
- ▶ Compound interest method
  - ▶ **Both simple and compound interest method**
- ▶ Accumulated method

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

Find x if  $3(x + 2) - 7 = 11$ .

- ▶ 2
- ▶ -4
- ▶ 6
- ▶ **4**

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

The value of x after solving the following linear equation is  
 $-2x + 6 = 4x - 2$

- ▶ 0
- ▶ 3
- ▶  $1/2$
- ▶  **$4/3$**

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

The sequence of payments at equal interval of time is called

- ▶ **Annuity**
- ▶ Accumulated value
- ▶ Discounted value
- ▶ None of these

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

Ali has improved his typing speed from 40w/m to 60w/m. The percentage improvement is

- ▶ 20%
- ▶ 150%

- ▶ 50%
- ▶ 30%

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

Markdown means a reduction from the

- ▶ Original cost price
- ▶ **Original sale price**
- ▶ Original Net price
- ▶ None of these

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

Markup is an amount added to -----

- ▶ get accumulated interest
- ▶ **a cost price while calculating a selling price.**
- ▶ The principal amount of the loan.
- ▶ Accumulation factor.

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

Interest calculated upon the principal amount added to the interest on it is called .....

- ▶ Simple interest
- ▶ **Compound interest**
- ▶ Annual interest per year
- ▶ Semi annual interest

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

Contribution Margin is the Rs. amount that is equal to -----

- ▶  **$S - VC$**
- ▶  $VC - S$
- ▶  $FC - VC$
- ▶  $VC - FC$

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

Reduction from original selling Price is called .....

- ▶ Loss

- ▶ List price
- ▶ Profit
- ▶ **Markdown**

**Question No: 14 ( Marks: 1 ) - Please choose one**

What will be the simple interest earned on an amount of Rs. 16,800 in 9 months at the

$$6\frac{1}{4}$$

rate of % p.a?

- ▶ **Rs. 787.50**
- ▶ Rs. 812.50
- ▶ Rs 860
- ▶ Rs. 887.50

**Question No: 15 ( Marks: 1 ) - Please choose one**

A percentage is a way of expressing a number as

□□□□□□

▶ **□a fraction of 100 .**

□□□□□□ ▶ □sum of number with 100.

□□□□□□ ▶ □multiplication of number with 100.

□□□□□□ ▶ □average of number with 100.

**Question No: 16 ( Marks: 1 ) - Please choose one**

12.5% of a circular plot as a fraction is expressed as **1/8**

$$\frac{5}{4}$$

□□□□□□ ▶ □

$$\frac{125}{1}$$

□□□□□□ ▶ □

$$\frac{8}{1}$$

□□□□□□ ▶ □

$$\frac{1}{8}$$

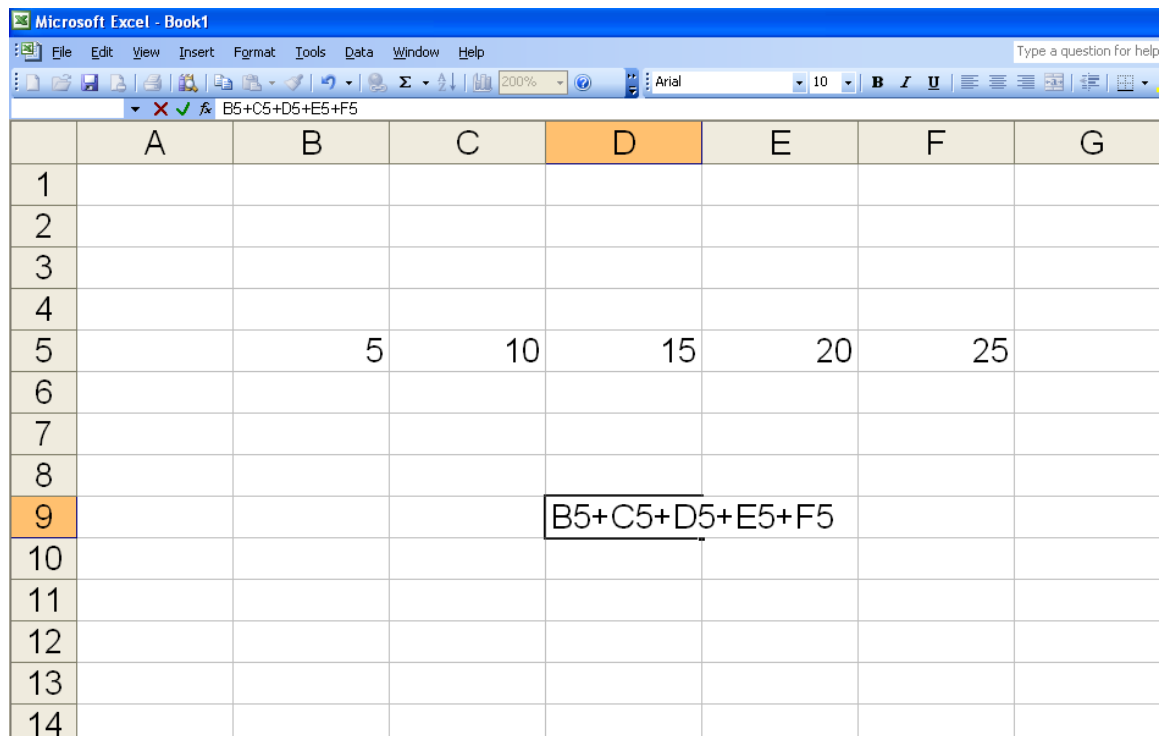
□□□□□□ ▶ □

**Question No: 17 ( Marks: 1 ) - Please choose one**

To convert a number in percent (%) from a fraction, we do which of the following operation:

- ☐ add by 100
- ☐ subtract by 100
- ☒ multiply by 100
- ☐ divide by 100

Question No: 18 ( Marks: 1 ) - Please choose one



|    | A | B | C  | D              | E  | F  | G |
|----|---|---|----|----------------|----|----|---|
| 1  |   |   |    |                |    |    |   |
| 2  |   |   |    |                |    |    |   |
| 3  |   |   |    |                |    |    |   |
| 4  |   |   |    |                |    |    |   |
| 5  |   | 5 | 10 | 15             | 20 | 25 |   |
| 6  |   |   |    |                |    |    |   |
| 7  |   |   |    |                |    |    |   |
| 8  |   |   |    |                |    |    |   |
| 9  |   |   |    | B5+C5+D5+E5+F5 |    |    |   |
| 10 |   |   |    |                |    |    |   |
| 11 |   |   |    |                |    |    |   |
| 12 |   |   |    |                |    |    |   |
| 13 |   |   |    |                |    |    |   |
| 14 |   |   |    |                |    |    |   |

What result will I get when press enter in cell D9?

- ☐ 75
- ☒ B5+C5+D5+E5+F5
- ☐ Error
- ☐ Cannot be determine

Question No: 19 ( Marks: 1 ) - Please choose one

---

Interest paid (earned) on only the original principal borrowed (lent) is often referred to as

▶ **simple interest**

▶ present value

▶ future value

▶ compound interest

**Question No: 20 ( Marks: 1 ) - Please choose one**

---

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix}$$

By multiplying the above two matrices we get the following result

$$C = \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}$$

▶

$$C = \begin{bmatrix} 9 \\ 7 \\ 17 \end{bmatrix}$$

▶

$$C = \begin{bmatrix} 9 \\ 12 \\ 17 \end{bmatrix}$$

▶

▶ **Given matrices cannot be multiplied**

**Question No: 21 ( Marks: 1 ) - Please choose one**

---

The formula for finding BEP in units is given by

▶ FC/VC

▶ **FC/(S-VC)**

▶ CM/FC

▶ CM/VC

**Question No: 22 ( Marks: 1 ) - Please choose one**

---

If A is a matrix whose determinant is zero then its inverse is given by:

- ▶ **zero**
- ▶ product of its diagonal entries
- ▶ infinite
- ▶ cannot be determined

**Question No: 23 ( Marks: 1 ) - Please choose one**

---

which of the following is linear equation

- ▶  **$2x-3y=-6$**
- ▶  $x+x+x$
- ▶  $520x^2y^2$
- ▶  $y^2-3=0$

**Question No: 24 ( Marks: 1 ) - Please choose one**

---

percentage of ratio of given number with standard number is.....

☐☐☐☐☐☐

▶ ☐100

☐☐☐☐☐☐☒ **Same number**

☐☐☐☐☐☐☐ ▶ ☐1000

☐☐☐☐☐☐☐ ▶ ☐10

**Question No: 25 ( Marks: 3 )**

---

Find the single discount rate that is equivalent to the series 18%, 12%, 9%.

Solution:

Let list price = 100

$$\begin{aligned}\text{Series trade discount: Netprice} &= \text{List price}(1-d_1)(1-d_2)\dots(1-d_n) \\ &= 100(1-.18)(1-.12)(1-.09) \\ &= 100(.82)(.88)(.91) \\ &= 100(.657) \\ &= 65.7\end{aligned}$$

$$\begin{aligned}\text{Single discount rate} &= 100 - 65.7 \\ &= 34.3\%\end{aligned}$$

**Question No: 26 ( Marks: 5 )**

---

Compute the amount of compound interest for Rs 3500 at 6% per annum for 2 years.

Solution:

$$S = P(1+R/100)^n$$

S= Money accrued after n years also called compound amount

P= Principle amount

R= Rate of interest

n= number of periods

$$S = 3500 (1 + 6/100)^2$$

$$= 3500 (1 + .06)^2$$

$$= 3500(1.06)^2$$

$$= 3500(1.1236)$$

$$S = 3933$$

$$\begin{aligned}\text{Compound interest} &= S - P \\ &= 3933 - 3500 \\ &= \mathbf{433}\end{aligned}$$

**Question No: 27 ( Marks: 10 )**

---

It is advised by the exam department of an institute to the assistant to develop a structure of marks obtained by three students in three different subjects for very quick view & analysis with the following instructions to be followed strictly as;

- a) marks obtained by EACH student in EVERY subject
- b) marks in EACH subject be obtained by EVERY student.

Also describe & justify the optimality of the structure. If such model is possible then suggest its further extension for numerous numbers of students.

**Solution:** The optimal method to solve this problem is by using **matrices**

Let the student names be:

1. Ahmed
2. Shahzad
3. Amjad

Let the subjects be:

1. Math
2. Chemistry

### 3. Physics

|         | Math | Chemistry | Physics |
|---------|------|-----------|---------|
| Ahmed   | 10   | 20        | 30      |
| Shahzad | 15   | 25        | 35      |
| Amjad   | 20   | 25        | 45      |

Now let's convert the above table into a matrix:

$$A = \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 35 \\ 20 & 25 & 45 \end{bmatrix}$$

**BENEFITS OF MATRICES:** The below benefits can help in further extension

- 1. Can handle large amount of data:**  
So w.r.t to the above question you can increase the no. of students & subjects as per your requirement and a
- 2. Analysis of data:**  
By using different matrix functions you can easily find out the respective averages and the grades of students.
- 3. Simple & Easy:**  
The above mentioned matrix structure is simple and easy to understand and the data can be easily converted from table to matrix
- 4. Easy to evaluate performance:**  
Its easy to find each students performance in a specific subject
- 5. Easy extension:**  
As matrices can handle large amounts of data so you just need to enter the data rest is done by matrices functions.

***Paper # 04***  
***Spring 2009 (session-4)***

***Time: 60 min***

***Marks: 40***

**Question No: 1 ( Marks: 1 ) - Please choose one**

---

A, B and C play cricket. A's runs are to B's runs and B's runs are to C's as 3:2. They get altogether 342 runs.

How many runs did A make?

▶ 162

▶ 108

▶ 72

▶ 112

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

The price of a P4 computer was \$12000 last year. What is the price of the computer for this year if the price is decreased by 10%?

▶ \$12000

▶ \$12500

▶ \$10800

▶ \$9000

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true?

▶ The mean is 103.

▶ The mean is 104.

- ▶ The median is 100.
- ▶ The median is 106.

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

Percentage is a ratio of given number with standard number:

- ▶ 100
- ▶ Same number
- ▶ 1000
- ▶ 10

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

The Virtual University of Pakistan is a growing distant learning university of Pakistan. The number of students in the university is increasing each year. The number of registered students in a course related to business math last year was 7113 and it is increased up to 9242 this year. What is the percentage increase in the number of students?

- ▶ 23%
- ▶ 30%
- ▶ 18%
- ▶ 42%

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

If A and B are any two matrices of order  $m \times n$  and  $p \times q$  respectively and  $m > n$  and  $p < q$ . What should be the condition on  $m, n, p, q$  for the product AB to hold?

- ▶  $n > p$
- ▶  $m \leq q$
- ▶  $q = p$
- ▶  $n = p$

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

The price at which investors buy or sell a share of stock at a given time is referred as

- ▶ Face value

▶ Market value

- ▶ Accumulated value
- ▶ Earning value

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

Gross salary is also called Net salary

- ▶ True
- ▶ False

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

Can we subtract a positive integer k from the identity matrix?

- ▶ Yes
- ▶ No

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

Net price =

- ▶ List price +trade discount

▶ List price - trade discount

- ▶ List price /trade discount

- ▶ List price \* trade discount

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

Division. Symbol: / is a type of

- ▶ Comparison operators

- ▶ Concatenation operator

- ▶ Reference operators

▶ Arithmetic operators

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

Given : List Price =5500Rs , Discount = 850Rs, then Net Cost Price will be

▶ 6350 Rs

6350 Rs

▶ 5508 Rs

5508 Rs

▶ 5585 Rs

▶ 4650 Rs

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

To convert a number in percent (%) from a fraction, we do which of the following operation:

▶ add by 100

▶ subtract by 100

▶ multiply by 100

▶ divide by 100

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

If  $A$  is the matrix of dimension  $2 \times 3$  and  $I$  is the identity matrix of dimension  $3 \times 3$ . Then which of the following is true

▶  $AI = I$

▶  $AI = A$

▶  $A + I = A$

▶  $A + I = I$

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

If ratio of boys and girls in a class is 7: 5, which of the following can't be the total number of the students in the class?

▶ 36

▶ 50

▶ 60

▶ 120

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

SLN returns the straight-line depreciation of an asset for

▶ Zero period

▶ One period

▶ Two periods

▶ Three periods

**Question No: 17 ( Marks: 1 ) - Please choose one**

---

Assume you will work for the next 20 years and at the end of each year you will deposit \$20,000 into a savings account. At the end of that 20 year period you will start withdrawing your retirement living expenses from that account at the beginning of every year. This financial plan involves a \_\_\_\_\_ problem while you are working and a \_\_\_\_\_ problem while you are in retirement.

▶ a present value of a dollar, future value of a dollar

▶ a future value of an annuity due, a present value of an annuity due

▶ a future value of an annuity due, a present value of an ordinary annuity

▶ a future value of an ordinary annuity, a present value of an annuity due

▶ a future value of an ordinary annuity, a present value of an ordinary annuity

**Question No: 18 ( Marks: 1 ) - Please choose one**

---

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix}$$

By multiplying the above two matrices we get the following result

$$C = \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}$$



$$C = \begin{bmatrix} 9 \\ 7 \\ 17 \end{bmatrix}$$



$$C = \begin{bmatrix} 9 \\ 12 \\ 17 \end{bmatrix}$$



**► Given matrices cannot be multiplied**

**Question No: 19 ( Marks: 1 ) - Please choose one**

which of the following is linear equation

►  $2x-3y=-6$

►  $x+x+x$

►  $520x^2y^2$

►  $y^2 - 3 = 0$

**Question No: 20 ( Marks: 1 ) - Please choose one**

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

□□□□□□□▶ □4500 units

□□□□□□▶ 167 units

□□□□□□▶ 33 units

□□□□□□▶ 16 units

**Question No: 21 ( Marks: 2 )**

---

Find the Discount where price is 2000 and discount rate is 12%.

Solution:

Price = 2000

Discount rate = 12%

Discount = Price \* rate

Discount =  $2000 \times 12 / 100$

Discount = 240

**Question No: 22 ( Marks: 3 )**

---

Calculate compound interest earned on Rs. 2250 invested at 8% per annum for 6 years

Solution

compound interest =  $(1 - ((1.08)^6)) \times 2250 = 1320$

**Question No: 23 ( Marks: 5 )**

---

If the Basic salary of an employee is Rs 8000 and Allowances are Rs 4,000 . What are the cost on account of 20 casual and 10 sick leaves per year if normal working days per month are 26?

Solution:

Cost of casual leaves per year =  $\{18 / (26 \times 12)\} \times 12000 \times 12 = 8307.7$  Rs

Cost of Sick leaves per year =  $\{12 / (26 \times 12)\} \times 12000 \times 12 = 5538.5$  Rs.

**Question No: 24 ( Marks: 10 )**

---

The price of medicine inventory is Rs. 500,000.

The series discounts are 20%, 10%, 5%.

What is the single equivalent discount rate?

if Invoice was dated May 1<sup>st</sup> and 10% discount is offered if invoice is paid up to 10<sup>th</sup> May. What will be net payment for invoice value of Rs. 500,000 if paid up to 10<sup>th</sup> May.

Solution

List Price=500,000

$$D1=20\%$$

$$D2=10\%$$

$$D3=5\%$$

$$\text{Net price}=N= L (1-d1)(1-d2)(1-d3)$$

$$= 500,000(1-0.20)(1-0.10)(1-0.05)$$

$$= 342,000$$

Single discounted rate is given by

$$\text{Here we see } N=L(1-d)$$

$$=500000(1-31.6\%)$$

$$=342000$$

So we can say that .....**31.6%** is a single discounted rate

$$\text{Payment over 10}^{\text{th}} \text{ of may} = 500,000(1-10\%)$$

$$= 450,000$$

*Paper # 05*  
*Spring 2010 (unknown)*

*Time: 60 min*  
*Marks: 40*

Q1

: In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- (a) Provident Fund
- (b) Gratuity Fund**
- (c) Charity Fund
- (d) None of the above

Q2

The screenshot shows a Microsoft Excel window with a formula bar containing `=SUMIF(C5:C8,>110,D5:D8)`. The worksheet displays a table with two columns: 'List price' and 'Discounted Price'. The rows list items: Calendar, Pen, Diary, and Table Lamp. The 'List price' column contains values 100, 150, 500, and 400. The 'Discounted Price' column contains values 90, 130, 470, and 385. A formula `=SUMIF(C5:C8,>110,D5:D8)` is entered in cell D10, with a tooltip showing the syntax: `SUMIF(range, criteria, [sum_range])`. The status bar at the bottom shows 'NUM'.

|            | List price | Discounted Price |
|------------|------------|------------------|
| Calendar   | 100        | 90               |
| Pen        | 150        | 130              |
| Diary      | 500        | 470              |
| Table Lamp | 400        | 385              |

SUMIF = `=SUMIF(C5:C8,>110,D5:D8)`

SUMIF(range, criteria, [sum\_range])

What will be the answer of the SUMIF formula?

- (a) 1075
- (b) 985**
- (c) 855
- (d) 130

Q3

If Cost = Rs. 7060 and Markup Rate = 46.7% then Selling price equal to -----

- (a) Rs. 11253.05
- (b) Rs. 32970.2
- (c) Rs. 10357.02**
- (d) Rs. 9046.7

Q4

Find x if  $2x + 6 = 10$ .

- (a) 8
- (b) 2**
- (c) -2
- (d) -8

Q5

Which statement is true?

- (a) Social charges comprise the leaves, group insurance and medical.**
- (b) Social charges comprise House rent and conveyance allowance.
- (c) Social charges comprise gratuity and provident funds.
- (d) Social charges comprise employee's children's education, club membership only.

Q6

0.20% = -----

- (a) 0.2
- (b) 0.02
- (c) 0.002**
- (d) 0.0002

Q7

$0^1 = ?$

- (a) 0**
- (b) 1
- (c) Undefined
- (d) Infinity

Q8

Net Cost Price =

- (a) List price – Discount.**
- (b) Discount - List price
- (c) Discount x List price
- (d) List price / Discount

Q9

If a speed of a car is changed from 25km/h to 40km/h then the percentage change in its speed is?

- (a) 15%

- (b) 50%
- (c) 40%
- (d) 60%**

Q10

Ali has improved his typing speed from 40w/m to 60w/m. The percentage improvement is

- (a) 20%
- (b) 150%
- (c) 50%**
- (d) 30%

Q11

*Amount of discount is obtained as*

- (a) Percentage of Discount x List Price**
- (b) Percentage of Discount / List Price
- (c) Percentage of Discount - List Price
- (d) None of these

Q12

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- (a) =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1\*12, 20%, 1)
- (b) =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- (c) =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
- (d) =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1\*12 , 20%/12, 1)
- (e) None of these

Q13

Net loss = Number of unit's sale below BEP in unit's × .....per unit.

- (a) FC
- (b) vC
- (c) CM**
- (d) S

Q14.

Given list price of shirt = \$20 , Discount = 10% ,Net price will be

- (a) \$ 16
- (b) \$ 18**
- (c) \$ 20
- (d) \$ 22

Q15

Given Old Sale price = \$1.95 , Markdown rate =13.6% ,then \$Mark down will be

- (a) \$ 0.27**
- (b) \$ 0.72
- (c) \$ 0.66
- (d) \$ 0.92

Q16

The Break Even Point in units is 64 and Production Capacity = 320 units per period, then The break Even point as a % of capacity will be

- (a) 35%
- (b) 30%
- (c) 25%
- (d) 20%**

Q17

The Basic salary of an employee is Rs 17, 000.

What is the contribution of the company on account of gratuity?

Q18

Given FC = Rs. 8000 , VC = Rs.200, S= Rs.190 , CM =Rs. 10 , NI = Rs 8000.

Find units over BEP

Q19

Store A marked down \$ 20.64 an item to \$ 14.29

What is the \$. Markdown?

What is the %markdown

Q20

The matrix G represents the average score for each student on tests, quizzes, and homework. Tests are 50% of the grade, quizzes are 30% and homework is 20%.

$$\begin{array}{c}
 \text{Amy} \\
 \text{Bill} \\
 \text{Chou} \\
 \text{David} \\
 \text{Erica}
 \end{array}
 \begin{bmatrix}
 \text{T} & \text{Q} & \text{H} \\
 78 & 80 & 75 \\
 76 & 90 & 95 \\
 72 & 70 & 85 \\
 60 & 70 & 80 \\
 84 & 80 & 90
 \end{bmatrix}
 = G$$

- a. Write the vector P expressing the percentages that would be used to find the final grade for each student.
- b. Would GP or PG produce a matrix of the final grades? Also using matrices, determine the final grade for each student.

Q21

Write which of the given function is used for which of the below given tasks.

**Functions:**

***RATE***(total number of payment periods, payment made each period,pv,fv,type,guess)

***PPMT***(interest rate, period, total number of payment periods, pv, fv, type)

**Paper # 06**  
**Spring 2010 (unknown)**  
**MTH302- Business Mathematics & Statistics**

**Time: 60 min**  
**Marks: 40**

Q1

Solve  $x - 16 = 12$  for  $x$ .

- (a) 4
- (b) 28**
- (c) -4
- (d) -28

Q2

The Virtual University of Pakistan is a growing distant learning university of Pakistan. The number of students in the university is increasing each year. The number of registered students in a course related to business math last year was 7113 and it is increased up to 9242 this year. What is the percentage increase in the number of students?

- (a) 23%**
- (b) 30%
- (c) 18%
- (d) 42%

Q3

Ali has bought his new motorcycle on installments. He will pay Rs. 2000 for 30 months. If the bike currently cost Rs. 54000 what percent of current cost he will pay more after 30 months.

- (a) 6%
- (b) 15%
- (c) 11.2%**
- (d) 10%

Q4

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

- (a) 5,000
- (b) 10,000**
- (c) zero
- (d) 15,000

Q5

The Basic salary of an employee is Rs 7,000 and the allowances are Rs 3000. Then the social charges will be -----.

- (a) Rs 2030

**(b) Rs 2900**

(c) Rs 2500

(d) Rs 2150

Q6

$0^1 = ?$

**(a) 0**

(b) 1

(c) Undefined

(d) Infinity

Q7

If a and b are any two integers such that  $a > b$  then  $a - b$  is

**(a) positive**

(b) negative

(c) zero

(d) cannot be told

Q8

If a birthday cake is cut into eight equal parts then what percent of the cake is a single pie?

(a) 20%

(b) 15.75%

**(c) 12.5%**

(d) 11%

Q9

A Matrix is a ..... of numbers

**(a) Rectangular array**

(b) Triangular array

(c) Linear array

(d) None of these

Q10

If a matrix has four column and 5 rows then its dimensions are

(a) 20

(b)  $4 \times 5$

**(c)  $5 \times 4$**

(d)  $5 \times 5$

Q11

Markup is an amount added to -----

(a) get accumulated interest

**(b) a cost price while calculating a selling price.**

(c) The principal amount of the loan.

(d) Accumulation factor.

Q12

In Excel there different types of operators are

(a) 3

- (b) 4
- (c) 5
- (d) 12

Q13

Assume you will work for the next 20 years and at the end of each year you will deposit \$20,000 into a savings account. At the end of that 20 year period you will start withdrawing your retirement living expenses from that account at the beginning of every year. This financial plan involves a \_\_\_\_\_ problem while you are working and a \_\_\_\_\_ problem while you are in retirement.

- (a) a present value of a dollar, future value of a dollar
- (b) a future value of an annuity due, a present value of an annuity due
- (c) a future value of an annuity due, a present value of an ordinary annuity
- (d) a future value of an ordinary annuity, a present value of an annuity due**
- (e) a future value of an ordinary annuity, a present value of an ordinary annuity

Q14

Which function calculates your monthly payment?

- (a) PMT**
- (b) NPER
- (c) PV
- (d) all of these choices

Q15

which of the following is linear equation

- (a)  $2x-3y=-6$**
- (b)  $x+x+x$
- (c)  $520x^2y^2$
- (d)  $y^2-3=0$

Q16

The point at which no profit is made and no losses are incurred is said to be

- (a) Fixed Cost point
- (b) Contribution Margin
- (c) Break even point**
- (d) Contribution Rate

Q17

If Variable Cost = Rs.157 and Sale= Rs.185, then find the value of contribution margin

Q18

If the basic salary of an employee Rs. 14,000 and allowances are Rs. 9,000 then what is the taxable income of employee?

Q19

Given the matrices

$$A = \begin{bmatrix} 0 & 9 \\ 2 & -3 \\ -1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 8 & 1 \\ -7 & 0 \\ 4 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 3 \\ -2 & 5 \\ 10 & -6 \end{bmatrix}$$

Compute

$$3A + 2B - \frac{1}{2}C$$

Q20

At what rate Rs. 4000 double it in 4 year.

Q21

A 50-inch segment is divided into three parts whose lengths have the ratio 2 : 3 : 5. What is the length of the longest part?.

## *Paper # 07*

*Spring 2011*

*MTH302- Business Mathematics & Statistics (Session - 2)*

**Question No: 1 ( Marks: 1 ) - Please choose one**

Principal remains constant through out the agreement period in:

**Compound interest**

Annuity

Simple interest

Nominal interest

**Question No: 2 ( Marks: 1 ) - Please choose one**

A square matrix A, has an inverse if,

A

= 0

**A**

= 0

A

0

A

= 0

**Question No: 3 ( Marks: 1 ) - Please choose one**

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

**I=PRT**

I=P+R+T

I=PT

$$I=P(R+T)$$

**Question No: 4 ( Marks: 1 ) - Please choose one**

If A and B are any two matrices of order  $m \times n$  and  $p \times q$  respectively and  $m > n$  and  $p < q$   
What should be the condition on  $m, n, p, q$  for the product AB to hold?

$n > p$

**$m \leq q$**

$q = p$

$n = p$

**Question No: 5 ( Marks: 1 ) - Please choose one**

Total Provident Fund added to the employee's fund is ----- of the basic salary.

**1 / 11 th**

2 / 11 th

9.09 %

9.99%

**Question No: 6 ( Marks: 1 ) - Please choose one**

2 6 1 9

A B 3 4 3 3, then AB is -----

20 36

25 30

20 36

35 30

20 36

15 39

20 36

45 39

**Question No: 7 ( Marks: 1 ) - Please choose one**

The negative of the negative of an odd number is

negative

**positive**

the odd number

does not exist

**Question No: 8 ( Marks: 1 ) - Please choose one**

The correct relation among the %markup on cost, cost price and selling price is

Selling Price = Cost price + (Cost price  $\times$  %Markup on sale)

**Selling Price = Cost price + (Cost price × %Markup on cost)**

Cost Price = selling price + (Cost price × %Markup on cost)

None of these

**Question No: 9 ( Marks: 1 ) - Please choose one**

If I is an identity matrix then it must also be a

**rectangular matrix**

row matrix

column matrix

scalar matrix

**Question No: 10 ( Marks: 1 ) - Please choose one**

The determinant of a square matrix is -----

**Matrix**

Number

Only a positive number

Only a negative number

**Question No: 11 ( Marks: 1 ) - Please choose one**

In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

Provident Fund

**Gratuity Fund**

Stock yield

Interest

**Question No: 12 ( Marks: 1 ) - Please choose one**

Returns the cumulative interest paid on a loan between start period and end period is .....

CUMIPMT

CUMPRINC

FV

PV

**Question No: 13 ( Marks: 1 ) - Please choose one**

In Excel there different types of operators are

3

**4**

5

12

**Question No: 14 ( Marks: 1 ) - Please choose one**

The text concatenation operator is used to

include “:” and “,”

calculate exponentiation: ^

**combine two text strings**

make comparisons.

**Question No: 15 ( Marks: 1 ) - Please choose one**

Division. Symbol: / is a type of

Comparison operators

Concatenation operator

Reference operators

**Arithmetic operators**

**Question No: 16 ( Marks: 1 ) - Please choose one**

12.5% of a circular plot as a fraction is expressed as

5

4

125

1

8

1

1

8

**Question No: 17 ( Marks: 1 ) - Please choose one**

Assume you will work for the next 20 years and at the end of each year you will deposit \$20,000 into a savings account. At the end of that 20 year period you will start withdrawing your retirement living expenses from that account at the beginning of every year. This financial plan involves a \_\_\_\_\_ problem while you are working and a \_\_\_\_\_ problem while you are in retirement.

a present value of a dollar, future value of a dollar

a future value of an annuity due, a present value of an annuity due

a future value of an annuity due, a present value of an ordinary annuity

**a future value of an ordinary annuity, a present value of an annuity due**

a future value of an ordinary annuity, a present value of an ordinary annuity

**Question No: 18 ( Marks: 1 ) - Please choose one**

Which of the following is the correct formula for calculating the contribution margin

**S-VC**

VC-S

S-FC  
FC-S

**Question No: 19 ( Marks: 1 ) - Please choose one**

Which is the correct syntax for the determinant of a matrix given by the following Array

=DETERM(B4:E7)  
=MDTERM(B4:E7)  
=**MDETERM(B4:E7)**  
=MDETERM(B4;E7)

**Question No: 20 ( Marks: 1 ) - Please choose one**

Which function calculates depreciation at the same amount each year over an asset's useful life?

**DB**  
SLN  
DDB  
All of above

**Question No: 21 ( Marks: 2 )**

An article is sold for Rs.990 causes 10% loss on the selling price, find its cost price.  
Answer  
1089

**Question No: 22 ( Marks: 3 )**

After a markup of 30% on sales a watch sells for Rs. 225. What is its cost price?  
Answer  
29.25

**Question No: 23 ( Marks: 5 )**

Cindy used part of an insurance settlement to purchase an ordinary annuity that would pay her \$5,000 each six months for 10 years. How much did the annuity cost if the interest rate is 10% compounded semiannually?(discounted value)

**Question No: 24 ( Marks: 10 )**

A stylish pencils that was bought for Rs. 200 was marked up 16% of the sale price. When the pencils went out of production, they were marked down 14%. What was the sale price after the 14% markdown?

***Paper # 08***  
***Spring 2011***  
***MTH302- Business Mathematics & Statistics (Session - 3)***

**Question No: 1 ( Marks: 1 ) - Please choose one**

What will be the answer of the SUMIF formula?

- 1075
- 985
- 855
- 130

**Question No: 2 ( Marks: 1 ) - Please choose one**

If you invest some amount at an interest rate of 8%, then at the end of 9 years.  
What will be the value of Accumulation Factor?

- 12.736
- 12.487
- 12.965
- 12.856

**Question No: 3 ( Marks: 1 ) - Please choose one**

If Cost = Rs. 7060 and Markup Rate = 46.7% then Selling price equal to -----

---

- Rs. 11253.05
- Rs. 32970.2**
- Rs. 10357.02
- Rs. 9046.7

**Question No: 4 ( Marks: 1 ) - Please choose one**

The margin is the difference between the selling price and the \_\_\_\_\_.

- Profit
- Revenue
- Cost**

tax

**Question No: 5 ( Marks: 1 ) - Please choose one**

Sale price is the selling price less the \_\_\_\_\_

- Markdown**
- Revenue
- Cost
- Profit

**Question No: 6 ( Marks: 1 ) - Please choose one**

Principal remains constant through out the agreement period in:

**Compound interest**

Annuity

Simple interest

Nominal interest

**Question No: 7 ( Marks: 1 ) - Please choose one**

Monthly installments of a leased car are calculated by using the techniques of:

Simple interest method

**Compound interest method**

Annuity

Hit & trial method

**Question No: 8 ( Marks: 1 ) - Please choose one**

Sum of annuity is an accumulated amount of:

Original payments and discount

**Original payments and interest**

Interest and the payment yet not paid

Compound interest

**Question No: 9 ( Marks: 1 ) - Please choose one**

Find x if  $3x + 7 - x = 17$ .

**5**

4

12

-5

**Question No: 10 ( Marks: 1 ) - Please choose one**

If

**Original Price**

= Rs. 3120 and

**Markdown Rate = 6.3 %**

then the

**Sale Price**

is equal

to

**2933.44**

3033.2

2910.4

3316.56

**Question No: 11 ( Marks: 1 ) - Please choose one**

Which statement is true?

**Social charges comprise the leaves, group insurance and medical.**

Social charges comprise House rent and conveyance allowance.

Social charges comprise gratuity and provident funds.

Social charges comprise employee's children's education, club membership only.

**Question No: 12 ( Marks: 1 ) - Please choose one**

Which of the following percent is of the basic salary?

Conveyance Allowance

**2.5%**

45%

5%

None of these

**Question No: 13 ( Marks: 1 ) - Please choose one**

If a and b are any two integers such that  $a > b$  then  $a - b$  is

**positive**

negative

zero

cannot be told

**Question No: 14 ( Marks: 1 ) - Please choose one**

The short cut method of showing a number is multiplied by itself is called

Superscript

Subscript

Exponent

**None of these**

**Question No: 15 ( Marks: 1 ) - Please choose one**

Dimension of a matrix means

Column space

Row space

**Order of the matrix**

None of these

**Question No: 16 ( Marks: 1 ) - Please choose one**

In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

Provident Fund

**Gratuity Fund**

Stock yield

Interest

**Question No: 17 ( Marks: 1 ) - Please choose one**

If  $x : y = y : 1$ , then

**$X = y$**

$X = y^2$

$$X = y / x$$

$$X = -y$$

**Question No: 18 ( Marks: 1 ) - Please choose one**

If the basic salary of an employee is 8, 000 Rs. allowances are 6000 Rs. and total cost of leaves per year is equal to 28280 Rs. then the cost of leaves as percent of gross salary is

**29.46%**

16.83%

22.23%

28.28%

**Question No: 19 ( Marks: 1 ) - Please choose one**

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

4500 units

**167 units**

33 units

16 units

**Question No: 20 ( Marks: 1 ) - Please choose one**

The Break Even Point in units is 64 and Production Capacity = 320 units per period, then The break Even point as a % of capacity will be

35%

30%

25%

**20%**

**Question No: 21 ( Marks: 2 )**

If a company is currently trading at Rs. 300 a share and earnings over the last 12 months were Rs. 10.5 per share, then find the price earning ratio for the stock

**Question No: 22 ( Marks: 3 )**

Given the matrices

Compute

A Matrix is a rectangular array of numbers. The plural of matrix is matrices.

Matrices are usually represented with capital letters such as Matrix A, B, C.

For example .

**Question No: 23 ( Marks: 5 )**

Find the interest earned on an annuity due if payments of \$2,000 per year are made for 15 years into an account paying 5% compounded annually.(Accumulated Value)

**Question No: 24 ( Marks: 10 )**

A,B and C, each of them working alone can complete a job in 6, 8 and 12 days respectively. If all three of them work together to complete a job and earn \$ 2340, what ill be share of each earnings?

Join:

***Paper # 09***  
***Spring 2011***  
***MTH302- Business Mathematics & Statistics (Session - 4)***

**Question No: 1 ( Marks: 1 ) - Please choose one**

---

If the basic salary is Rs 6500 and the social charges are Rs 1820, what percentage of basic salary are the social charges?

- ▶ 27 %
- ▶ **28 %**
- ▶ 29 %
- ▶ 30 %

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

The price at which a business purchases merchandise is called the

- ▶ List
- ▶ **Cost**
- ▶ Investment
- ▶ Exchange rate

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

A reduction of the amount due on an invoice is called a \_\_\_\_\_ .

- ▶ **Trade discount**
- ▶ Net discount
- ▶ Cash discount
- ▶ Unearned discount

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

If **Selling Price** = Rs. 5890 and **Markup Rate** = 59.5 % then the **Cost Price** is equal to

- ▶ Rs. 3229.70
- ▶ 3692.79
- ▶ **2385.45**
- ▶ 3504.55

**Question No: 5 ( Marks: 1 ) - Please choose one**

0.20% = -----

- ▶ 0.2
- ▶ 0.02
- ▶ **0.002**
- ▶ 0.0002

**Question No: 6 ( Marks: 1 ) - Please choose one**

What percentage is 10% of 85?

- ▶ ☐☐☐
- ▶ ☐☐☐
- ▶ ☐☐☐
- ▶ **None of these**

**Question No: 7 ( Marks: 1 ) - Please choose one**

If a matrix has four column and 5 rows then its dimensions are

- ☐☐☐☐☐☐ ▶ ☐☐☐
- ☐☐☐☐☐☐ ▶ ☐☐☐☐
- ☐☐☐☐☐☐ ▶ **☐☐☐☐☐☐**
- ☐☐☐☐☐☐ ▶ ☐☐☐☐

**Question No: 8 ( Marks: 1 ) - Please choose one**

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ⌚ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1\*12, 20%, 1)
  - ⌚ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
  - ⌚ **=AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
  - ⌚ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1\*12 , 20%/12,
- 1)
- ⌚ None of these

**Question No: 9 ( Marks: 1 ) - Please choose one**

The break even point is a point when -----

- ▶ Revenue is greater than cost.
- ▶ Revenue is less than cost.
- ▶ **Revenue is exactly equal to cost.**
- ▶ None of the above.

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$6,000. The useful life of the asset is 4 years. The depreciation is being calculated for the first year, and there are 12 months in the first year.

- ▶ =DB (10000, 6000, 4, 1, 12)
- ▶ **=DB (10000, 6000, 4, 12, 1)**
- ▶ =DB (6000, 10000, 4, 12, 1)
- ▶ =DB (10000, 6000, 4, 1, 1)

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

The text concatenation operator is used to

- ▶ include “:” and “,”
- ▶ calculate exponentiation: ^
- ▶ **combine two text strings**
- ▶ make comparisons.

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

Which of the following formulas is the correct formula for calculating selling price

- ▶ Cost price (1+Cost price × %Markup on cost)
- ▶ **Cost price (1+ %Markup on cost)**
- ▶ Cost price (Cost price × %Markup on cost)
- ▶ Cost price (Cost price + Cost price × %Markup on cost)

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%

▶ 25%

▶ **20%**

▶ 10%

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

▶ 4500 units

▶ **167 units**

▶ 33 units

▶ 16 units

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

$3x^2 + 5x - 7$  is ----- expression.

▶ **Monomial**

▶ Binomial

▶ Trinomial

▶ Linear

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

If there is a change of -30% in the price of an item, what does the negative sign show?

▶ **The price is decreasing.**

▶ The price has low rate of change.

▶ None of the above.

▶ The price is increasing.

**Question No: 17 ( Marks: 1 ) - Please choose one**

---

The price-earning ratio is calculated as.....

▶ **Market value per share/Earning per share (EPS)**

▶ Earning per share (EPS)/Market value per share

▶ Market value per share+Earning per share (EPS)

▶ Market value per share-Earning per share (EPS)

**Question No: 18 ( Marks: 1 ) - Please choose one**

---

Net current asset value per share (NCAVPS) is calculated as.....

- ▶ No. of shares outstanding/(Current assets-total liabilities)
- ▶ **(Current assets-total liabilities)/No. of shares outstanding**
- ▶ No. of shares outstanding+(Current assets-total liabilities)
- ▶ No. of shares outstanding-(Current assets-total liabilities)

**Question No: 19 ( Marks: 1 ) - Please choose one**

---

What shall be contribution margin if sale price is Rs.50 and variable cost is Rs.30?

- ▶ Rs.80
- ▶ **Rs.20**
- ▶ Rs.30
- ▶ none of these

**Question No: 20 ( Marks: 1 ) - Please choose one**

---

For multiplication of two matrices A and B

- ▶ **The no. of rows of A must be equal to columns of B**
- ▶ The no. of rows of B must be equal to no. columns of A
- ▶ The no. of columns of A must be equal to no. rows of B
- ▶ None of these

**Question No: 21 ( Marks: 2 )**

---

The price of a sandwich maker is Rs 2000. The company offers a 25% trade discount. Find trade discount amount.

**Solution**

Sandwich Price                      Rs.2, 000

|                       |                 |
|-----------------------|-----------------|
| Trade Discount Rate   | 25%             |
| Trade Discount Amount | =2,000*25/100   |
|                       | = <b>Rs.500</b> |

**Question No: 22 ( Marks: 2 )**

A firm sells its new product at Rs. 65 per unit. If the variable cost is Rs 40, find the contribution margin.

### Solution

|                     |                         |
|---------------------|-------------------------|
| Sale Price Per Unit | Rs.65                   |
| Variable Cost       | Rs.40                   |
| Contribution Margin | =S-VC                   |
|                     | =65-40                  |
|                     | = <b>Rs.25 per unit</b> |

**Question No: 23 ( Marks: 3 )**

Ratio of sales of Product X to sales of Product Y is 5:4. The sales of product X is forecasted at Rs. 200,000. What should be the Sales of product Y to maintain the ratio of sales between the two products?

### Solution

|                      |     |                     |
|----------------------|-----|---------------------|
| Ratio x to y         | 5:4 |                     |
| Sale of Product x    |     | Rs.200, 000         |
| Sale of product of y |     | =200,000 * 4/5      |
|                      |     | <b>=Rs.160, 000</b> |

**Question No: 24 ( Marks: 3 )**

A company sells its item at Rs. 125 per unit. If the variable cost is Rs 80, find the contribution rate.

### Solution

|                     |             |
|---------------------|-------------|
| Sale Price per unit | Rs.125      |
| Variable Cost       | Rs.80       |
| Contribution margin | =S-VC       |
|                     | =125-80     |
|                     | =Rs.45      |
| Contribution Rate   | =CM/S*100   |
|                     | =45/125*100 |

$$=36\%$$

**Question No: 25 (Marks: 5)**

Ahmed buys 10 apples for \$1. At what price should he sell a dozen apples if he wishes to make a profit of 25%?

**Solution**

|                            |                        |
|----------------------------|------------------------|
| Cost price                 | = \$1 for 10 Apples    |
| Cost price per apple       | = $1/10$               |
|                            | = \$0.1                |
| Cost price of dozen apples | = $0.1 \times 12$      |
|                            | = \$1.2                |
| Profit Margin              | 25%                    |
| Sale price                 | = $1.2 \times 125/100$ |
|                            | = \$1.5                |

**Question No: 26 ( Marks: 5 )**

Evaluate the sum of the years digits depreciation of an asset having the cost of Rs.10, 000 for a period of 5 years such that salvage value after 7 years is Rs.5000.

**Solution**

|                                   |                                  |
|-----------------------------------|----------------------------------|
| Cost Price                        | Rs.10, 000                       |
| Salvage value after 7 years       | Rs.5, 000                        |
| Depreciable Value                 | = Cost Price – Salvage value     |
|                                   | = $10,000 - 5,000$               |
|                                   | = 5,000                          |
| Sum of Years digit depreciation.  | = $7 + 6 + 5 + 4 + 3 + 2 + 1$    |
|                                   | = 28                             |
| 1st year depreciation             | = $7/28 \times 5000$ = Rs.1, 250 |
| 2 <sup>nd</sup> year depreciation | = $6/28 \times 5000$ = Rs.1, 071 |
| 3rd year depreciation             | = $5/28 \times 5000$ = Rs.893    |
| 4 <sup>th</sup> year depreciation | = $4/28 \times 5000$ = Rs.714    |
| 5 <sup>th</sup> year depreciation | = $3/28 \times 5000$ = Rs.536    |